



ECx00G&EG800G Series

QuecOpen(SDK) PWM Development Guide

LTE Standard Module Series

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Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

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About the Document

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1 Introduction

Quectel ECx00G family and EG800G series modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS. It is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see **document [1]**.

This document is applicable to QuecOpen® solution based on CSDK build environment. It introduces the development of PWM in application layer, including PWM API, development example and debugging of Quectel ECx00G family and EG800G series modules.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
ECx00G	EC200G-CN
	EC600G Series
	EC800G-CN
-	EG800G Series

1.2. Module PWM Channels

ECx00G family and EG800G series modules support a maximum of 16 PWM channels. For details about the specific channels opened to different modules, contact Quectel Technical Support.

2 PWM API

2.1. Header File

ql_pwm.h, the header file of PWM API, is in the `\components\ql-kernel\inc` directory. Unless otherwise specified, all header files mentioned in this document are in this directory.

2.2. API Overview

Table 2: API Overview

Function	Description
<i>ql_pwm_open()</i>	Opens a PWM channel.
<i>ql_pwm_close()</i>	Closes a PWM channel.
<i>ql_pwm_enable()</i>	Enables a PWM channel and configures pulse period and duty cycle.
<i>ql_pwm_disable()</i>	Disables a PWM channel.

2.3. API Description

2.3.1. ql_pwm_open

This function opens a PWM channel.

- **Prototype**

```
ql_errcode_pwm ql_pwm_open(ql_pwm_sel pwm_sel)
```

- **Parameter**

pwm_sel:

[In] PWM channel. See **Chapter 2.3.1.1** for details.

- **Return Value**

See **Chapter 2.3.1.2** for details.

2.3.1.1. ql_pwm_sel

The enumeration of PWM channel types:

```
typedef enum
{
    QL_PWM_0,
    QL_PWM_1,
    QL_PWM_2,
    QL_PWM_3,
    QL_PWM_4,
    QL_PWM_5,
    QL_PWM_6,
    QL_PWM_7,
    QL_PWM_8,
    QL_PWM_9,
    QL_PWM_10,
    QL_PWM_11,
    QL_PWM_12,
    QL_PWM_13,
    QL_PWM_14,
    QL_PWM_15,
    QL_PWM_MAX
}ql_pwm_sel
```

- **Member**

Member	Description
QL_PWM_0	PWM0 channel.
QL_PWM_1	PWM1 channel.
QL_PWM_2	PWM2 channel.
QL_PWM_3	PWM3 channel.

QL_PWM_4	PWM4 channel.
QL_PWM_5	PWM5 channel.
QL_PWM_6	PWM6 channel.
QL_PWM_7	PWM7 channel.
QL_PWM_8	PWM8 channel.
QL_PWM_9	PWM9 channel.
QL_PWM_10	PWM10 channel.
QL_PWM_11	PWM11 channel.
QL_PWM_12	PWM12 channel.
QL_PWM_13	PWM13 channel.
QL_PWM_14	PWM14 channel.
QL_PWM_15	PWM15 channel.
PWM_MAX	Invalid channel.

NOTE

For details about the usage of the PWM channel, please contact Quectel Technical Support to obtain the GPIO configuration document for the corresponding module.

2.3.1.2. ql_errcode_pwm

PWM result codes indicate whether the function is executed successfully. An error cause is returned if the execution fails. The enumeration of result codes of PWM:

```
typedef enum
{
    QL_PWM_SUCCESS = QL_SUCCESS,

    QL_PWM_EXECUTE_ERR           = 1|QL_PWM_ERRCODE_BASE,
    QL_PWM_INVALID_PARAM_ERR,
    QL_PWM_FUNC_SET_ERR,
    QL_PWM_ACQUIRE_ERR,
    QL_PWM_START_ERR,
    QL_PWM_STOP_ERR,
    QL_PWM_REPEAT_OPEN_ERR,
```

```

    QL_PWM_REPEAT_CLOSE_ERR
}ql_errcode_pwm

```

● Member

Member	Description
QL_PWM_SUCCESS	Successful execution.
QL_PWM_EXECUTE_ERR	Failed execution.
QL_PWM_INVALID_PARAM_ERR	Parameter error.
QL_PWM_FUNC_SET_ERR	Failed to configure PWM.
QL_PWM_ACQUIRE_ERR	Failed to get PWM information.
QL_PWM_START_ERR	Failed to enable PWM.
QL_PWM_STOP_ERR	Failed to stop PWM.
QL_PWM_REPEAT_OPEN_ERR	Repeating opening a PWM channel error.
QL_PWM_REPEAT_CLOSE_ERR	Repeating closing a PWM channel error.

2.3.2. ql_pwm_close

This function closes a PWM channel.

● Prototype

```
ql_errcode_pwm ql_pwm_close(ql_pwm_sel pwm_sel)
```

● Parameter

pwm_sel:

[In] PWM channel. See **Chapter 2.3.1.1** for details.

● Return Value

See **Chapter 2.3.1.2** for details.

2.3.3. ql_pwm_enable

This function enables a PWM channel and configures pulse period and duty cycle.

● Prototype

```
ql_errcode_pwm ql_pwm_enable(ql_pwm_sel pwm_sel, ql_pwm_cfg_s *pwm_cfg)
```

● Parameter

pwm_sel:

[In] PWM channel. See **Chapter 2.3.1.1** for details.

pwm_cfg:

[In] Parameters that need to be configured to enable PWM. See **Chapter 2.3.3.1** for details.

● Return Value

See **Chapter 2.3.1.2** for details.

2.3.3.1. ql_pwm_cfg_s

The structure of PWM parameter configuration:

```
typedef struct
{
    ql_pwm_clk clk_sel;
    uint16_t   prescaler;
    uint16_t   period;
    uint16_t   duty;
} ql_pwm_cfg_s
```

● Parameter

Type	Parameter	Description
<i>ql_pwm_clk</i>	<i>clk_sel</i>	PWM clock source. See Chapter 2.3.3.2 for details.
uint16_t	<i>prescaler</i>	Prescaler.
uint16_t	<i>period</i>	Value that automatically loads the counter.
uint16_t	<i>duty</i>	Count value corresponding to pin at high level.

NOTE

1. Output frequency $f = \text{Clock source frequency} / (\text{prescaler} + 1) / (\text{period} + 1)$.
2. Duty cycle = $\text{duty} / \text{period}$.

2.3.3.2. ql_pwm_clk

The enumeration of PWM clock source:

```
typedef enum
{
    QL_CLK_XTAL,
    QL_CLK_XTAL_LP,
    QL_CLK_RC26M,
    QL_CLK_GNSS_PLL_133M,
    QL_CLK_GNSS_PLL_198M,
    QL_CLK_APLL_200M,
    QL_CLK_APLL_250M,
    QL_CLK_RTC_32K,
} ql_pwm_clk
```

● Member

Member	Description
<i>QL_CLK_XTAL</i>	Currently not supported.
<i>QL_CLK_XTAL_LP</i>	Internal clock source (default). Frequency: 26 MHz.
<i>QL_CLK_RC26M</i>	RC oscillator clock source. Frequency: 26 MHz.
<i>QL_CLK_GNSS_PLL_133M</i>	Currently not supported.
<i>QL_CLK_GNSS_PLL_198M</i>	Currently not supported.
<i>QL_CLK_APLL_200M</i>	Currently not supported.
<i>QL_CLK_APLL_250M</i>	Currently not supported.
<i>QL_CLK_RTC_32K</i>	Currently not supported.

2.3.4. ql_pwm_disable

This function disables a PWM channel.

- **Prototype**

```
ql_errcode_pwm ql_pwm_disable(ql_pwm_sel pwm_sel)
```

- **Parameter**

pwm_sel:

[In] PWM channel. See **Chapter 2.3.1.1** for details.

- **Return Value**

See **Chapter 2.3.1.2** for details.

3 Development Example and Debugging

This chapter introduces how to use the above API for PWM development and simple debugging in application layer.

3.1. Development Example in Application Layer

3.1.1. Example Description

led_cfg_demo.c, the example file, is provided in the `\components\ql-application\peripheral` directory of Quectel ECx00G and EG800G family QuecOpen SDK. The example file contains configuration and usage of PWM, including opening, closing, disabling and enabling a PWM channel. The entry function is *ql_ledcfg_app_init()* as shown below.

The example sets the flashing mode of the network indicator light according to the network state of the current (U)SIM card.

```
void ql_ledcfg_app_init(void)
{
    QLOSStatus err = QL_OSI_SUCCESS;

    err = ql_rtos_task_create(&ledcfg_task, 1024, APP_PRIORITY_NORMAL, "ql_ledcfgdemo", ql_ledcfg_demo_thread, NULL, 1);
    if( err != QL_OSI_SUCCESS )
    {
        QL_LEDcfgDEMO_LOG("led config demo task created failed");
    }
}
```

Figure 1: Flashing Mode of the Network Indicator Light

3.1.2. Example Auto-start

The above example is enabled by default in `ql_init_demo_thread()` as shown below.

```
static void ql_init_demo_thread(void *param)
{
    ...QL_INIT_LOG("init_demo_thread_enter, param:0x%x", param);
    #if 0
    ...ql_gpio_app_init();
    ...ql_gpioint_app_init();
    #endif
    ...ql_ledcfg_app_init();
    #ifdef QL_APP_FEATURE_AUDIO
    ...ql_audio_app_init();
    #endif
    #ifdef QL_APP_FEATURE_LCD
    ...ql_lcd_app_init();
    #endif
    ...ql_nw_app_init();
    ...//ql_datacall_app_init();
    ...ql_osi_demo_init();
}
```

Figure 2: Example Auto-start

3.2. PWM Debugging

3.2.1. Preparation

PWM of ECx00G family and EG800G series modules can be debugged on Quectel LTE OPEN EVB. You can view information of the example after capturing the AP logs with ArmLogel. For details on log capture, see [document \[2\]](#).

NOTE

Insert a (U)SIM card before you debug the example.

3.2.2. Debugging

`ql_ledcfg_app_init()` automatically starts after the module is booted. Through log information, you can see the flashing frequency of the network indicator light. At the same time, you can see the actual on-off state of the NET_STA indicator light of the corresponding PWM channel on the EVB, and you can also use an oscilloscope to measure the corresponding pin signals to see if the settings are correct.

The log debugging information is shown below:

```
[QUEC/I] quec_ledcfg_event_send:89 led config id=82020001, param1=0, param2=0,...
[QUEC/I] quec_ledcfg_handler:265 get event: 0x82020001/0x00000000/0x00000000/0...
[QUEC/I] quec_ledcfg_event_send:89 led config id=82020001, param1=0, param2=0,...
[QUEC/I] quec_ledcfg_handler:265 get event: 0x82020001/0x00000000/0x00000000/0...
[QOPN/I] ql_ledcfg_demo_thread:81 led config demo thread enter, param 0x0
[QUEC/I] quec_ledcfg_handler:265 get event: 0x82020002/0x00000000/0x00000000/0...
[QOPN/I] _ledcfg_demo_cb:61 led config net_status is 0, net_type is 0
[QUEC/I] quec_ledcfg_handler:322 led config info get
[QOPN/I] ql_ledcfg_demo_thread:111 led config slow1 twinkle [0]
```

Figure 3: Log Debugging Information

NOTE

Because the pins used in this example may be configured in other examples, make sure that only this example runs when you debug PWM.

4 Appendix References

Table 3: Related Documents

Document Name
[1] Quectel_ECx00G&EC600U&EG800G_Series_QuecOpen(SDK)_Quick_Start_Guide
[2] Quectel_ECx00G&EG800G_Series_QuecOpen(SDK)_Log_Capture_Guide

Table 4: Terms and Abbreviations

Abbreviation	Description
(U)SIM	(Universal) Subscriber Identity Module
AP	Access Point
APB	Advanced Peripheral Bus
API	Application Programming Interface
EVB	Evaluation Board
GPIO	General-Purpose Input/Output
IoT	Internet of Things
LED	Light Emitting Diode
LTE	Long-Term Evolution
PC	Personal Computer
PWM	Pulse Width Modulation
RTOS	Real-Time Operating System
SDK	Software Development Kit

UART	Universal Asynchronous Receiver/Transmitter
USB	Universal Serial Bus
